7. Explain the Levels/Tasks of NLP.

1) INTRODUCTION

Natural Language Processing (NLP) is a hierarchical pipeline where raw text is processed through sequential stages, from lexical analysis to pragmatic interpretation. Each level transforms the output of the previous stage into a more abstract representation, ultimately enabling the system to understand and generate human-like language.

2) LEVELS OF NLP (HIERARCHY)

5. Pragmatic Level → *Understanding context, intent, and world knowledge to interpret meaning beyond literal syntax.*

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4. Semantic Level → *Deriving the literal meaning of sentences and identifying relationships between entities.*

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3. Syntactic Level → *Analyzing grammatical structure and parsing sentences into constituent trees.*

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2. Morphological Level → *Analyzing word structure, inflections, and deriving base forms (lemmas).*

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1. Lexical Level → *Tokenization and identifying basic units of language (words, punctuation).*

3) DESCRIPTION OF LEVELS / TASKS

Level / Task	Goal	Main Tasks	Example
1. Lexical Level	Break raw text into manageable units and identify their basic properties.	• Tokenization • Part-of-Speech (POS) Tagging • Normalization (lowercasing, removing punctuation)	**Input :** "The quick brown fox jumps." **Output :** [The/DET] [quick/ADJ] [brown/ADJ] [fox/NN] [jumps/VBZ] [./PUNCT]
2. Morphological Level	Analyze word formation and reduce words to their root forms.	• Stemming • Lemmatization • Inflection analysis	**Input :** "jumps", "running", "better" **Output :** "jump", "run", "good"
3. Syntactic Level	Determine the grammatical structure of the sentence.	• Parsing (Constituent/Dependency) • Chunking • Identifying Subject-Verb-Object relations	**Input :** "The quick brown fox jumps." **Output :** (S (NP (DT The) (JJ quick) (JJ brown) (NN fox)) (VP (VBZ jumps)) (PUNCT .))
4. Semantic Level	Extract the literal meaning and logical relationships.	• Word Sense Disambiguation (WSD) • Named Entity Recognition (NER) • Semantic Role Labeling (SRL)	**Input :** "The bank is closed." **Output :** Entity: 'bank' (Financial Institution), Relation: 'closed' (State)
5. Pragmatic Level	Interpret intent, context, and implied meaning.	• Intent Detection • Coreference Resolution • Discourse Analysis	**Input :** "Can you pass the salt?" **Output :** Intent: Request for action (not a question about ability)

4) NLP AS A PIPELINE OF TASKS

• Sequential Flow: Output of Level N is Input to Level N+1 •

Key Points ■

- NLP is a bottom-up process: Structure is built from primitives (tokens) to complex meaning (pragmatics).
- Each level adds a layer of abstraction; skipping a level leads to incorrect interpretation.
- Pragmatics is the highest level, requiring external world knowledge and context, not just linguistic rules.
- Modern NLP often combines these levels using deep learning, but the conceptual hierarchy remains valid for understanding.

Example (End to End) ■

Input : "I saw the man with the telescope."

1. Lexical: Tokens: [I/PRP] [saw/VBD] [the/DT] [man/NN] [with/IN] [the/DT] [telescope/NN]

2. Morphological: Lemmas: [I] [see] [the] [man] [with] [the] [telescope]

3. Syntactic: Parse: (S (NP I) (VP (VBD saw) (NP (DT the) (NN man) (PP (IN with) (NP (DT the) (NN telescope))))))

4. Semantic: Ambiguity: Did 'I' have the telescope, or did 'the man' have it?

5. Pragmatic: Context: If 'I' am an astronomer, likely 'I' had the telescope. If 'the man' is a spy, likely 'he' had it.

KEY DEFINITIONS, LAWS & FORMULAS (MUST-MEMORIZE)

• **Tokenization**: Splitting raw text into discrete units (words, punctuation).

Formula / Rule: Text -> [Token1, Token2, ..., TokenN]

• **Lemmatization**: Mapping inflected words to their dictionary form using a lexicon.

Formula / Rule: Word -> Lemma (Base Form)

• **Parse Tree**: Hierarchical representation of syntactic structure.

Formula / Rule: S -> NP VP

4. EXHAUSTIVE QUESTIONS & ANSWERS • (6 Questions)

Q1. Define the Lexical Level in NLP. [2 Marks • Short Concept] ■■■ V.IMP

- Lowest level of NLP processing.
- Involves tokenization and POS tagging.
- Identifies basic units of language.

■ ***Examiner Tip:*** *Mention 'Tokenization' and 'POS Tagging' explicitly.*

Q2. What is the difference between Stemming and Lemmatization? [2 Marks • Short Concept] ■■■ V.IMP

- Stemming: Rule-based, cuts off suffixes (may result in non-words).
- Lemmatization: Dictionary-based, finds the correct base form (lemma).
- Lemmatization is more accurate but computationally expensive.

■ ***Examiner Tip:*** *Use the example 'running' -> 'runn' vs 'run' to illustrate.*

Q3. Explain the Syntactic and Semantic levels of NLP with examples. [5 Marks • Medium Explanation] ■■■■ IMP

- Syntactic: Grammatical structure, parsing trees.
- Semantic: Meaning, word sense, relationships.
- Example: 'Bank' (river vs. financial).
- Syntactic parsing identifies 'bank' as a noun; Semantic identifies which type of noun.

■ ***Examiner Tip:*** *Draw a small parse tree for the syntactic part to score full marks.*

Q4. Describe the Pragmatic Level of NLP. [5 Marks • Medium Explanation] ■■■■ IMP

- Highest level of NLP.
- Involves context, intent, and world knowledge.
- Handles ambiguity and implied meaning.
- Example: 'Can you pass the salt?' is a request, not a question.

■ ***Examiner Tip:*** *Emphasize 'Intent' and 'World Knowledge' as key differentiators.*

Q5. Explain the complete hierarchy of NLP levels with a detailed example. [10 Marks • Long Explanation]
■■■■■ **V.IMP**

• Introduction to NLP hierarchy.
• Level 1: Lexical (Tokenization, POS).
• Level 2: Morphological (Stemming, Lemmatization).
• Level 3: Syntactic (Parsing, Chunking).
• Level 4: Semantic (WSD, NER, SRL).
• Level 5: Pragmatic (Intent, Context).
• End-to-end example: 'I saw the man with the telescope.'
• Conclusion: Importance of sequential processing.

■ ***Examiner Tip:*** *Use a table or bullet points for each level. Include a clear example for each level.*

Q6. Compare and contrast the Syntactic and Semantic levels of NLP. [10 Marks • Comparative Analysis]
■■■■■ **V.IMP**

• Syntactic: Structure-focused, rule-based, parse trees.
• Semantic: Meaning-focused, context-dependent, logical forms.
• Syntactic handles grammar; Semantic handles truth and meaning.
• Example: 'Colorless green ideas sleep furiously' (Syntactically correct, Semantically nonsensical).
• Interdependence: Syntax provides structure for Semantic analysis.

■ ***Examiner Tip:*** *Use the 'Colorless green ideas' example to highlight the difference.*

5. HIGH-YIELD MEMORY TRICKS & MNEMONICS

■ **L-M-S-S-P: Lexical -> Morphological -> Syntactic -> Semantic -> Pragmatic (Bottom to Top).**

■ **Think of it as a pyramid: Tokens at the base, Intent at the peak.**

■ **Syntax = Grammar, Semantics = Meaning, Pragmatics = Context.**